Mudit Arora

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Education

University of California, Santa Cruz Dec 2025
Master of Science, Artificial Intelligence | GPA: 3.86 *Santa Cruz, CA*
- **Courses**: Machine Learning, Deep Learning, Natural Language Processing, LLM Agents, Generative AI, Conversational AI, Multimodal AI, Data Science, AI Agents, AI in Games

Arizona State University May 2024
Bachelor of Science, Computer Science | GPA: 3.77 *Tempe, AZ*
- **Awards/Honors**: Magna Cum Laude, Dean's List, New American University Scholar, SUN Award
- **Courses**: Artificial Intelligence, Data Structures & Algorithms, Software Engineering, Computational Biology

Technical Skills

Languages: Python, R, TypeScript, JavaScript, C++, MATLAB, Java, Swift
Frameworks/Libraries: PyTorch, TensorFlow, scikit-learn, Keras, MLX, NLTK, Pandas, NumPy, LangChain, AutoGen, matplotlib, OpenCV, React, Next.js, Node.js, HTML/CSS
Tools & Technologies: GCP, Azure, AWS (S3), Git, MySQL, Docker, Postman, Hugging Face, Ollama, Make.com
Domains: Machine Learning, Deep Learning, Voice AI, Data Sceince, NLP, LLM, RAG, Generative AI, AI Agents
Certifications: Technical Interview Prep (CodePath), Web Development (CodePath)

Experience

Member of Technical Staff Jul 2026 – Present
Synhawk *San Francisco, CA*
- First research hire in the San Francisco office; established engineering workflows and infrastructure for a team that scaled to **6 researchers**.
- Designed and built an end-to-end synthetic data generation pipeline using **Python, Airflow, Cloudflare R2**, producing **750 samples weekly** across **audio, image, and video** for model training and evaluation.
- Integrated **AI agents** and **custom evaluation metrics** into the pipeline, enabling fully automated generation and quality checks with **zero manual intervention**.
- Built an internal analytics dashboard in **JavaScript** for daily data visibility, reducing manual review time by **35%**.

Deep Learning Researcher May 2025 – Feb 2026
Uniphore *Palo Alto, CA*
- Generated **500+ synthetic conversations** based on ReAct & Pre-Act frameworks across domains (Banking, Travel, etc.) for a flexible dialog agent capable of multi-turn reasoning and dynamic tool calling on normal, happy, and unhappy paths.
- Trained a series of **Qwen Small Language Models** using **Supervised Fine Tuning (SFT)** to generalize tool use and tuned **Reinforcement Learning with Verifiable Rewards (RLVR)** on Action, Tool, and Semantic Rewards.
- Achieved **50% and 66.7% success rates** on seen and unseen domains respectively (**12x improvement** over base models).

AI Software Engineer Intern May 2025 – Sept. 2025
CRED *San Francisco, CA*
- Worked on internal automation tools for PMs to take notes, review transcripts, and identify key problems and solutions, **saving up to 2 hours** per workflow.
- Designed an AI bug-fixing agent for writing fixes, Q&A, and reviewing bugs, **saving developers up to 4 hours**.
- Crafted features for CRED's Chrome Extension using LLMs for intelligent preprocessing and semantic chunking to scrape webpage data, achieving **84% accuracy** in benchmark test, and enhanced the process to retrieve real-time data to users.
- Optimized CRED's AI Slack Bot using OCR models, **reducing wait time by 15%**.

Machine Learning Researcher Aug. 2023 – May 2024
Mayo Clinic *Tempe, AZ*
- Fine-tuned **Google's T5**-based LLM using **Python**, **PyTorch**, and **scikit-learn**, achieving **87% accuracy** in extracting social determinants of health from clinical notes and predicting patient readmission within 30 days.
- Helped hospitals **reduce admission rates, lowering care costs** in patient care management & improve clinical decisions.

Projects

Slug Meditate | ***CruzHacks 2025 Winner***
- Built a VR meditation web app pipeline utilizing **Google's Gemini API** to transform user text prompts into AI-generated images (**Imagen 3**) and videos (**Veo 2**), transformed into 3D scene mapping (**Gaussian Splatting**) with AI-generated music (**MusicFX**), rendered in VR Immersion (**Niantic Studio by 8th Wall**).
- Achieved an **87% success rate** in rendering immersive experiences across 14+ unique user prompts.

Blindspot | ***Smallest.ai Hackathon Runner Up***
- Engineered an accessibility-focused iOS app for blind/visually impaired users featuring on-device ML-powered object/scene detection (**CoreML + VisionKit**) and voice AI (**Smallest.ai + Grok**).
- Delivered real-time performance with **<2s response latency**, reducing navigation uncertainty.

EduMUSE
- Developed an AI-powered tutoring system using **CrewAI multi-agent architecture** that transforms PDF study materials into personalized learning experiences with automated summaries, quiz generation, and podcast-style audio content.
- Reduced study material processing time by **68%**.